

다변수 미적분학 (BS101-02)

학기	2026년 2학기
담당교수	안흥주
이수구분	공통필수
학점	3.0
강의시간	월14:00-16:00(E7 - L31), 수14:00-16:00(E7 - L31)
강의실	E7 218 (교수실)
소속	기초학부

교과개요

Must download an attached file Introduce students in engineering, physics, computer science, and related fields to the areas of applied mathematics relevant to solving practical problems. In particular, the following topics will be covered: Fourier analysis, linear PDEs such as the wave equation, heat equation, Laplace equation, the theory of complex functions including the calculus of complex functions of complex variables, and their applications to residue integral and potential theory.

교과목표

Learning Objectives will be given at attached files (weekly schedule part)

주차별 강의계획

1주차:What is a vector calculus, Cartesian coordinates, vectors, inner product, lines, and planes
2주차:cylindrical, spherical coordinates, matrices and operations, elementary row operations, existence of system of linear equations, and determinant of 3×3 matrix
3주차:Cross product and its meaning, quadric surfaces,
4주차:Two variables function, limits, graphs, continuity, and partial derivatives,
5주차:Higher order partial derivatives, differentiability and tangent planes
6주차:Chain rule, Taylor's theorem for two variables function
7주차:Hesse criterion, Lagrange multiplier
8주차:Mid Exam
9주차:Double, triple integral, iterated integral, and compute multiple integrals
10주차:Jacobian and change of variables and applications of change of integral
11주차:Line integral, reparametrization of path and Green Theorem I
12주차:Green Theorem II, Path independent line integral, conservative field, closed field and scalar potential
13주차:Parametrization of surface, surface integral
14주차:Circulation of vector field along path, flux of vector field across surface, Stokes Theorem, Divergence Theorem
15주차:Meaning of Divergence Theorem and Stokes Theorem, applications to Maxwell's equations
16주차:Final Exam

수업방법

Lecture-based instruction. Students are encouraged to discover problems on their own, understand the historical process of problem solving, and use lectures when necessary to solve problems. Theoretical learning is combined with computational approaches using Matlab.

학습윤리

Attendance is the most basic behavior that confirms the sincerity of learning activities, and the attendance verification requires students to be honest. Unethical behavior in attendance is strictly prohibited, such as proxy attendance and unauthorized early departure. Writing and submitting assignments is the act of presenting the results of one's learning, which can be enhanced by including the learner's own thoughts and expressions in the assignment. Unethical behavior in writing and submitting assignments includes falsification and alteration, plagiarism, and duplicate submissions, which are strictly prohibited. Cooperative learning is the act of acquiring diverse knowledge and creating new knowledge and should always be based on your own sincere efforts. Free-riding is the act of taking advantage of the hard work of others without putting in a fair amount of effort, which is unethical because it harms the people involved. Tests are designed to measure students' academic performance and are based on fair assessment. Cheating can be done by asking someone else to take the test or by using someone else as a substitute.

교과정책(방침)

Must attend recitation (연습시간). Recitation consists of working through Matlab, solving exercises, asking questions, etc. and may be facilitated by a TA. Up to 5 unexcused absences allowed. More than 6 unexcused absences (20%) may result in a lower grade than the original. Same as above for attendance for recitation, but allow up to 3 absences (20% excluding exam week). Absolute evaluation of the 02 class only (02반 대상 절대평가). Failure to attend midterm and final exams will result in an F grade regardless of score (there is no make-up exam). Anything below 20 points out of 100 (20 is not an absolute score, but with some variation depending on the tests) is an F. In the event of an unavoidable withdrawal by the student, credit may reasonably be granted, taking into account the points earned by the student, as evidenced by clear documentation.

준비물 및 기타

Personal laptop with Matlab installed (Matlab program can be installed freely inside university)